

Chapter 8

Regression and Correlation Analysis

Learning Objectives

- Use simple linear regression for building empirical models of engineering and scientific data.
- Understand how the method of least squares is used to estimate the parameters in a linear regression model.
- Test statistical hypotheses and construct confidence intervals on regression model parameters.
- Use the regression model to make a prediction of a future observation and construct an appropriate prediction interval on the future observation.
- Apply the correlation model.

8.0 Regression and Correlation Analysis

8.1 Introduction

Many problems in engineering and science involve exploring the relationships between two or more variables. **Regression analysis** is a statistical technique that is very useful for these types of problems. For example, in a chemical process, suppose that the yield of the product is related to the process-operating temperature. Regression analysis can be used to build a model to predict yield at a given temperature level. This model can also be used for process optimization, such as finding the level of temperature that maximizes yield, or for process control purposes. Other examples are, studying the relationship between blood pressure and age, the concentration of an injected drug and heart rate etc.

Regression analysis is concerned with the study of the dependence of one variable, the dependent variable, on one or more other variables, the independent or explanatory variables with a view to estimating and predicting the (population) mean or average of the former (dependent) in terms of the known or fixed (in repeated sampling) values of the latter (independent).

Very often in practice, a relationship is found to exist between two (or more) variables and one wishes to express this relationship in mathematical form by determining an equation connecting the variables.

Correlation analysis, on the other hand, is concerned with measuring the strength of the relationship between variables. When we compute measures of correlation from a set of data, we are interested in the degree of the *correlation* between variables.

8.2 The Regression Model

In the typical regression problem, the researcher has available for analysis a sample of observations from some real or hypothetical population. Based on the result of his analysis of the sample data, he is interested in reaching decisions about the population from which the sample is presumed to have been drawn. It is important that the researcher understand the nature of the population in which he is interested.

In the simple linear regression model two variables X and Y , are of interest. The variable X is usually referred to as the independent variable, while the other variable, Y is called the dependent variable; and we speak of the regression of Y on X .

The following are the assumptions underlying the simple linear regression model.

- (i) Values of the independent variable X are fixed.
- (ii) The variable X is measured without error.
- (iii) Values Y are normally distributed
- (iv) The Y values are statistically independent.
- (v) The variances of the subpopulations of Y are all equal.

$$V(Y|x) = V(\alpha + \beta x + e) = V(\alpha + \beta x) + V(e) = 0 + \sigma^2 = \sigma^2$$

- (i) The means of the subpopulations of Y all lie on the same straight line, this is the assumption of linearity.

$$E(Y|x) = \mu_{Y|x} = \alpha + \beta x \quad (1.0)$$

These assumptions may be summarized by means of the following equation which is called the simple linear regression model because it has only one independent variable or **regressor**:

$$y = \alpha + \beta x + e \quad (1.1)$$

where α and β (slope and intercept) are called population regression coefficients, e is called the **error term** with **mean zero** and variance σ^2 . The random errors corresponding to different observations are also assumed to be uncorrelated random variables.

The results of n observations of the set of random variables X and Y can be summarized by drawing a scatter diagram. A straight line passing closely to the points may be drawn. The main problem arises when the points do not all lie exactly on the straight line, but simply form a cloud of points around it. Thus, it may be possible by guess work to draw quite a number of lines each of which will appear to be able to explain the relationship between X and Y . We shall consider finding a best fit line. Such a line will then be used as a model relating the random variable Y with the random variable X .

Suppose that we have n pairs of observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. The following figure shows a typical scatter plot of observed data and a candidate for the estimated regression line.

The estimates of α and β should result in a line that is (in some sense) a “best fit” to the data. The German scientist Karl Gauss proposed estimating the parameters α and β in Equation 1.1 to minimize the sum of the squares of the deviations in the diagram. We call this criterion for estimating the regression coefficients the **method of least squares**. Using Equation 1.1, we may express the n observations in the sample as

$$y_i = \alpha + \beta x_i + e_i$$

$$i = 1, 2, \dots, n$$

and the sum of the squares of the deviations of the observations from the true regression line is

$$L = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2$$

(1.2)

The least squares estimators of α and β , say $\hat{\alpha}$ and $\hat{\beta}$, must satisfy

$$\frac{\partial L}{\partial \alpha} = -2 \sum_{i=1}^n (y - \hat{\alpha} - \hat{\beta}x_i) = 0$$

(1.3)

$$\frac{\partial L}{\partial \beta} = -2 \sum_{i=1}^n (y - \hat{\alpha} - \hat{\beta}x_i) x_i = 0$$

Simplifying these two equations yields

$$n\hat{\alpha} + \hat{\beta} \sum_{i=1}^n x_i = \sum_{i=1}^n y_i$$

(1.4)

$$\hat{\alpha} \sum_{i=1}^n x_i + \hat{\beta} \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i$$

Equations 1.4 are called the least squares normal equations. The solution to the normal equations results in the least squares estimators $\hat{\alpha}$ and $\hat{\beta}$.

The least squares estimates of the intercept and slope in the simple linear regression model are

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} \quad (1.5)$$

$$\begin{aligned} \hat{\beta} &= \frac{\sum_{i=1}^n x_i y_i - \frac{\left(\sum_{i=1}^n x_i\right)\left(\sum_{i=1}^n y_i\right)}{n}}{\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n}} \\ &= \frac{n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i\right)\left(\sum_{i=1}^n y_i\right)}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i\right)^2} \end{aligned}$$

(1.6)

Equation 1.6 can also be written as

$$\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2}$$

The estimated regression line is therefore

$$\hat{y} = \hat{\alpha} + \hat{\beta}x$$

Alternatively

8.3 Method of Least Squares

We shall now find a and b , the estimates of α and β so that the sum of the squares of the residuals is a minimum. The residual sum of squares is often called the Sum of Squares of Errors (SSE) about the regression line. This minimisation procedure for estimating the parameter is called the “methods of least squares”. Hence we shall find a and b so as to minimise

$$SSE = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - a - bx_i)^2$$

Differentiating SSE with respect to a and b , we have

$$\begin{aligned} \frac{\partial(SSE)}{\partial a} &= -2 \sum_{i=1}^n (Y_i - a - bx_i) \\ \frac{\partial(SSE)}{\partial b} &= -2 \sum_{i=1}^n (Y_i - a - bx_i)x_i \end{aligned}$$

Setting the partial derivative equal to zero and rearranging the terms, we obtain the equation (called the normal equations)

$$na + b \sum_{i=1}^n x_i = \sum_{i=1}^n y_i \quad \dots (1)$$

$$a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i \quad \dots (2)$$

Solving for a and b from (1) and (2)

$$b = \frac{n \sum x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{n \sum x_i^2 - \left(\sum_{i=1}^n x_i \right)^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{SS_{xy}}{SS_{xx}}$$

$$a = \frac{\sum_{i=1}^n y_i - b \sum_{i=1}^n x_i}{n} = \bar{y} - b\bar{x}$$

Equations (1) and (2) can also be solved using matrices as:

$$\begin{bmatrix} n & \sum x \\ \sum x & \sum x^2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \sum y \\ \sum xy \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} n & \sum x \\ \sum x & \sum x^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum y \\ \sum xy \end{bmatrix}$$

Example 91

Assuming we have the following quantities $n = 8$, $\sum x = 140$, $\sum y = 382$, $\sum xy = 3870$, and $\sum x^2 = 3500$.

Solution

Using the above equation:

$$\begin{bmatrix} 8 & 140 \\ 140 & 3500 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 382 \\ 3870 \end{bmatrix}$$

Solving gives $a = 94.67$ and $b = -2.68$

Thus, the estimated regression line is given by:

$$= 94.67 - 2.68x$$

It may be noted that the least-squares line passes through the point $(\bar{x}, \bar{y})$ called the ‘Centroid’ or centre of gravity of the data. The slope b of the regression line is independent of the origin of coordinates. It is therefore said that b is invariable under the translation of axes.

Besides assuming that the regression of y and x is a linear function having the form $E(Y|X) = \alpha + \beta x$ we have made three further assumptions which may be summarised as follows:

1. **Normality:** We have assumed that each variable y_i has a normal distribution.
2. **Independence:** We have assumed that the variables $y_1, \dots, y_n$ are independent.
 1. **Homoscedasticity:-** We have assumed that the variable $y_1, \dots, y_n$ have the same variance σ^2 . This assumption is called the assumption of homoscedasticity. In general, it is said that random variables having the same variance are homoscedastic, and random variables having different variance are heteroscedastic.

Example 92

A team of professional mental health workers in a long-stay psychiatric hospital wished to measure the level of response of withdrawn patients to a program of remotivation therapy. A standard test was available for this purpose, but it was expensive and time-consuming to administer. To overcome this obstacle, the team developed a test that was much easier to administer. To test the usefulness of the new instrument for measuring the level of patient response, the team decided to examine the relationship between scores made on the new test and scores made on the standardized test. The objective was to use the new test if it could be shown that it was a good predictor of a patient’s score on the standardized test. The results are shown in the table below:

Patients' Scores on Standardized Test and New Test		
Patient Number	Score on New Test (X)	Score on Standardized Test (Y)
1	50	61
2	55	61
3	60	59
4	65	71
5	70	80
6	75	76
7	80	90
8	85	106
9	90	98
10	95	100
11	100	114

Obtain the estimates of the regression coefficients.

Example 93

A research on “Near Surface Characteristics of Concrete: Intrinsic Permeability”, presented data on compressive strength x and intrinsic permeability y of various concrete mixes and cures. Summary quantities are $n = 14$, $\sum x_i = 43$, $\sum y_i = 572$, $\sum x_i^2 = 157.42$, $\sum y_i^2 = 23,530$, and $\sum x_i y_i = 1697.80$. Assume that the two variables are related according to the simple linear regression model.

- Calculate the least squares estimates of the slope and intercept.
- Use the equation of the fitted line to predict what permeability would be observed when the compressive strength is $\bar{x} = 4.3$.
- Give a point estimate of the mean permeability when compressive strength is $\bar{y} = 3.7$.

Example 94

The following data were obtained from a study investigating the relationship between noise exposure and hypertension.

Y	1	0	1	2	5	1	4	6	2	3	5	4	6	8	4	5	7	9	7	6
X	60	63	65	70	70	70	80	90	80	80	85	89	90	90	90	90	94	100	100	100

- Fit the simple linear regression model using least squares.
- Find the predicted mean rise in blood pressure level associated with a sound pressure level of 85 decibels.

8.5 Correlation Analysis

Closely related but conceptually very much different from regression analysis is **correlation analysis**, where the primary objective is to measure the *strength or degree of linear association* between two variables. The **correlation coefficient** measures this strength of (linear) association. For example, we may be interested in finding the correlation between smoking and lung cancer; between scores on mathematics and fluid mechanics examinations, between high school grades and

college grades etc. In regression analysis, as already noted, we are not primarily interested in such a measure. Instead, we try to estimate the average value of one variable on the basis of the fixed values of another variable.

The population correlation coefficient between two random variables, X and Y is defined as

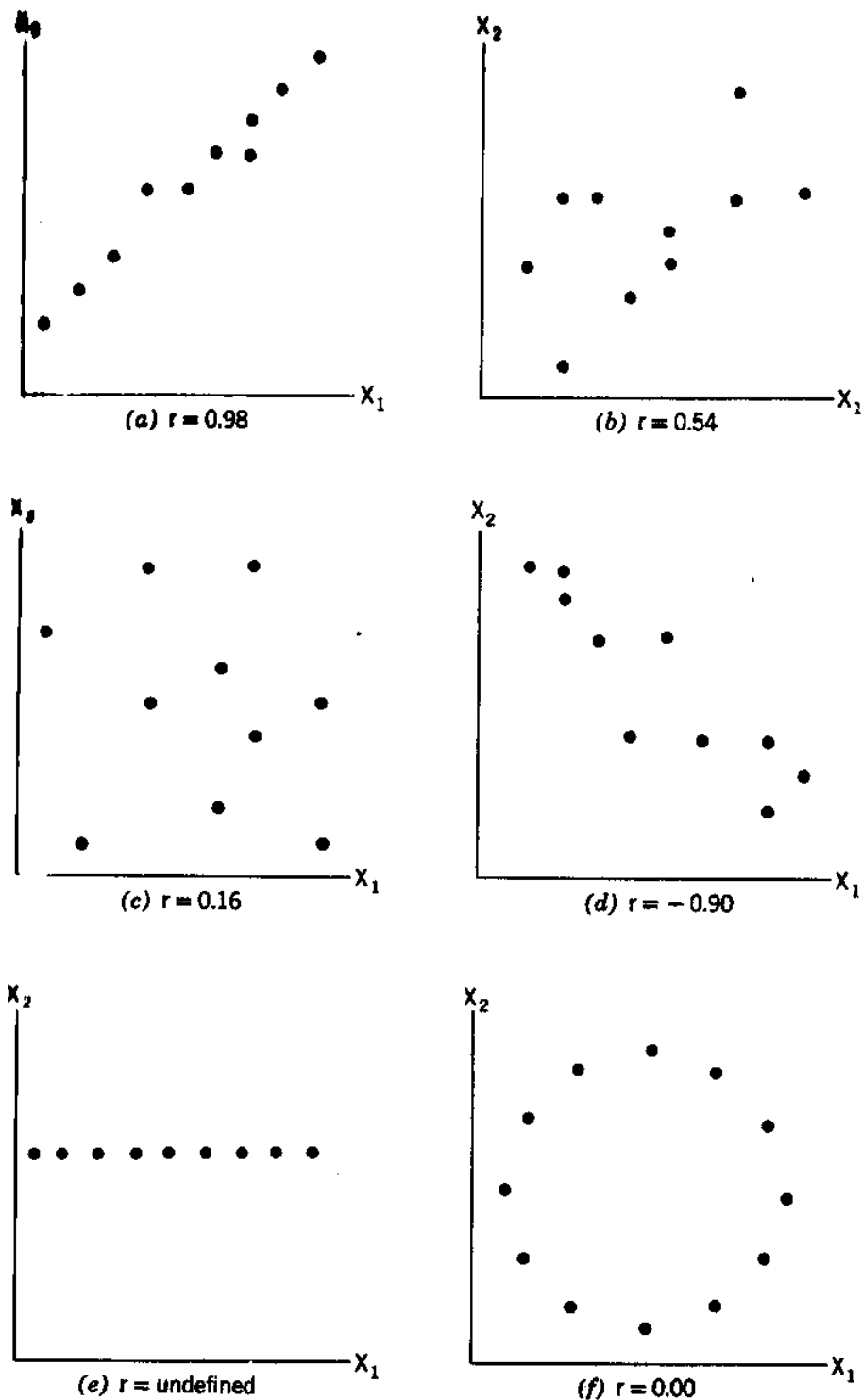
$$\rho = \frac{E[X - E(X)][Y - E(Y)]}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{\sigma_{XY}}{\sigma_X\sigma_Y}$$

where σ_{XY} is the covariance between variables X and Y , σ_X and σ_Y are the standard deviations of X and Y respectively. It is possible to draw inferences about the correlation coefficient ρ using its estimator, the sample correlation coefficient, r . “ r ” is the correlation coefficient between “ n ” pairs of observations whose values are (X_i, Y_i) and is given by

$$\begin{aligned} r &= \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sqrt{(\sum_{i=1}^n x_i^2 - n\bar{x}^2)(\sum_{i=1}^n y_i^2 - n\bar{y}^2)}} \\ &= \frac{n \sum_{i=1}^n x_i y_i - (\sum_{i=1}^n x_i)(\sum_{i=1}^n y_i)}{\sqrt{(n \sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2)(n \sum_{i=1}^n y_i^2 - (\sum_{i=1}^n y_i)^2)}} \end{aligned}$$

Properties of r :

1. It is symmetrical in nature (the two variables are treated symmetrically). That is, there is no distinction between the dependent and independent variables.
2. Both variables are assumed to be random.
3. It can be positive or negative, the sign depending on the sign of the term in the numerator which measures the sample co variation of the two variables.
4. It lies between the limits of -1 and +1; that is, $-1 \leq r \leq +1$.
5. If X and Y are independent, the correlation coefficient between them is zero but if $r=0$ it does not mean that the two variables are independent.

Figure 8.1: Fig. 6.2 Scatter plots with various r values**Testing hypothesis about the correlation coefficient.**

A test of the special hypothesis $\rho = 0$ versus an appropriate alternative is equivalent to testing $\beta = 0$ for the simple linear regression model. In doing this the t -distribution with $n-2$ degrees of freedom may be needed.

t

$$= \frac{b}{\sqrt{SS_{xx}}}$$

which can also be written as to test:

$$H_0 : \rho = 0$$

$$H_1 : \rho \neq 0$$

Example 95

Using the following data, test the hypothesis that there is no linear correlation among the variables that generated them; at 5% level of significance:

$$SS_{xx} = 0.11273 \quad SS_{yy} = 11,807,324,786 \quad SS_{xy} = 34,422,759.72$$

Solution

$$r = \frac{SS_{xy}}{\sqrt{SS_{xx}SS_{yy}}} = \frac{34422.75972}{\sqrt{(0.11273)(11807324786)}} = 0.9435$$

$$H_0 : \rho = 0$$

$$H_1 : \rho \neq 0$$

$$\alpha = 0.05 \quad df = (n - 2)$$

critical region: $t < -2.052$ and $t > 2.052$

$$t = \frac{0.9435(\sqrt{29-2})}{\sqrt{1-(0.9435)^2}} = 14.79$$

$$P < 0.0001$$

Decision:

Since $t > t_{0.025}(27)$ reject the hypothesis of no linear correlation.

More generally, if X and Y follow the bivariate normal distribution, it can be shown that quantity.

is a random variable that follows approximately the normal distribution with mean. and variance equal to $1/(n-3)$. There the procedure is to compute

$$\begin{aligned} z &= \frac{\sqrt{n-3}}{2} \left[\ln \left(\frac{1+r}{1-r} \right) - \ln \left(\frac{1+\rho_0}{1-\rho_0} \right) \right] \\ &= \frac{\sqrt{n-3}}{2} \ln \left[\frac{(1+r)(1-\rho_0)}{(1-r)(1+\rho_0)} \right] \end{aligned}$$

Example 96

Consider the immediate preceding example data, test the null hypothesis that $\rho = 0.9$ against the alternative that $\rho > 0.9$ at 5% level of significance.

Solution

$$H_0 : \rho = 0.9$$

$$H_1 : \rho > 0.9$$

Critical region: $Z > 1.645$

Decision: Since $Z < Z_{0.05}$ there is no evidence that the correlation coefficient is not equal to 0.9

In ordinary usage of this method, it is not necessary to use the formula for Z that corresponds to r values between 0.0 and 0.99. Tables contain fisher - Z values Z_f are available. In this case to test $H_0: \rho = \rho_0$ vs $H_1: \rho \neq \rho_0$

$Z =$ we have $Z =$

Critical region is $Z \leq -Z_{\alpha/2}$ and $Z \geq Z_{\alpha/2}$ where Z_f and μ_f are the fisher - Z values for r and ρ_0 respectively.

Example 97

The following data gave $X =$ the water content of snow on April 1 and $Y =$ the yield from April to July (in inches) on the Snake River watershed in Wyoming for 17 years.

x	y	X	y
23.1	10.5	37.9	22.8
32.8	16.7	30.5	14.1
31.8	18.2	25.1	12.9
32.0	17.0	12.4	8.8
30.4	16.3	35.1	17.4
24.0	10.5	31.5	14.9
39.5	23.1	21.1	10.5
24.2	12.4	27.6	16.1
52.5	24.9		

Estimate the correlation between X and Y .

Example 98

Two methods of measuring cardiac output were compared in 10 experimental animals with the following results

<i>Cardiac Output (l./min)</i>	
<i>Method I</i> x	<i>Method I</i> Y
0.8	0.5
1.0	1.2
1.3	1.1
1.4	1.3
1.5	1.1
1.4	1.8
2.0	1.6
2.4	2.0
2.7	2.4
3.0	2.8

Compute the sample correlation coefficient.

Example 99

A group of eight athletes ran a 400 metres race twice. The times in seconds were recorded as follows for each athlete.

<i>Runner</i>	
<i>1st Trial</i> x	<i>2nd Trial</i> Y
48.4	48.0
51.2	54.3
48.6	49.4
49.5	48.4
51.6	54.0
49.3	47.2
50.8	51.8
49.7	50.3

Calculate the correlation coefficient between these two trials.

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